CDS6224 | Statistical Data Analysis
Assignment 1 : Descriptive Statistics G Data Visualization Analysis | Term 2510

Weightage: **30%**
(Marks will be deducted by 10% for each day late. No submission will be accepted after being 3 days late)
Submission Deadline: **11th May 2025, 11.59PM**
Submission Platform: **eBwise**

Objective:

This assignment challenges your ability to apply statistical techniques to real-world data. You are required to not only perform computations using R but also to interpret and communicate insights effectively, justify your choice of statistical tools, and reflect critically on your findings.

Dataset:

Any dataset (you may find in Kaggle/UCI Repository etc.) with at least **two numerical variables** and **one categorical variable**.
Kaggle dataset:https://www.kaggle.com/datasets/iamsouravbanerjee/heart-attack-prediction-dataset?resource=download

The heart attack risk prediction dataset has 8763 records, offering 26 features on patient health. I choose cholesterol and heart rate as my quantitative variable for this assignment, heart attack risk and categorical variable. I choose only the first 5000 records as my research data. I changed the column 'Heart Rate' into 'Heart_Rate' so that it is easier to code.

Justify why you selected the dataset and what questions you aim to explore.
Cardiovascular disease is a major global health concern, which makes this dataset highly relevant and meaningful for real-world applications.

1. *How does cholesterol relate to heart attack?*
2. *Which is a better indicator of heart disease risk: cholesterol or heart rate?*

Assignment Tasks G Mark Distribution:

Part A: Exploratory Descriptive Analysis (10 marks)

1. Choose any two quantitative variables from your selected dataset. Provide a detailed summary: mean, median, mode, standard deviation, variance, range, quartiles, IQR and CV.

Cholesterol

Mean	*260.2376475*
Median	*257*

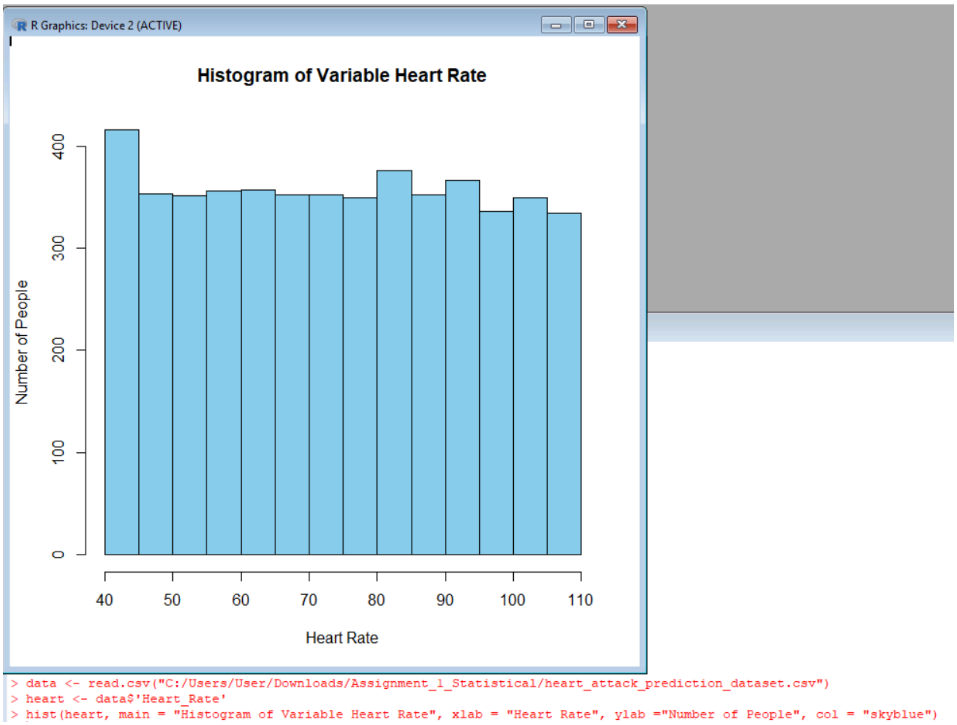
Mode	*235*
Standard Deviation	*80.82348984*
Variance	*6531.129761*
Range	*400-120=280*
Quartiles	*Q1: 193* *Q2: 257* *Q3: 330*
IQR	*137*
CV	*31.06%*

Heart Rate

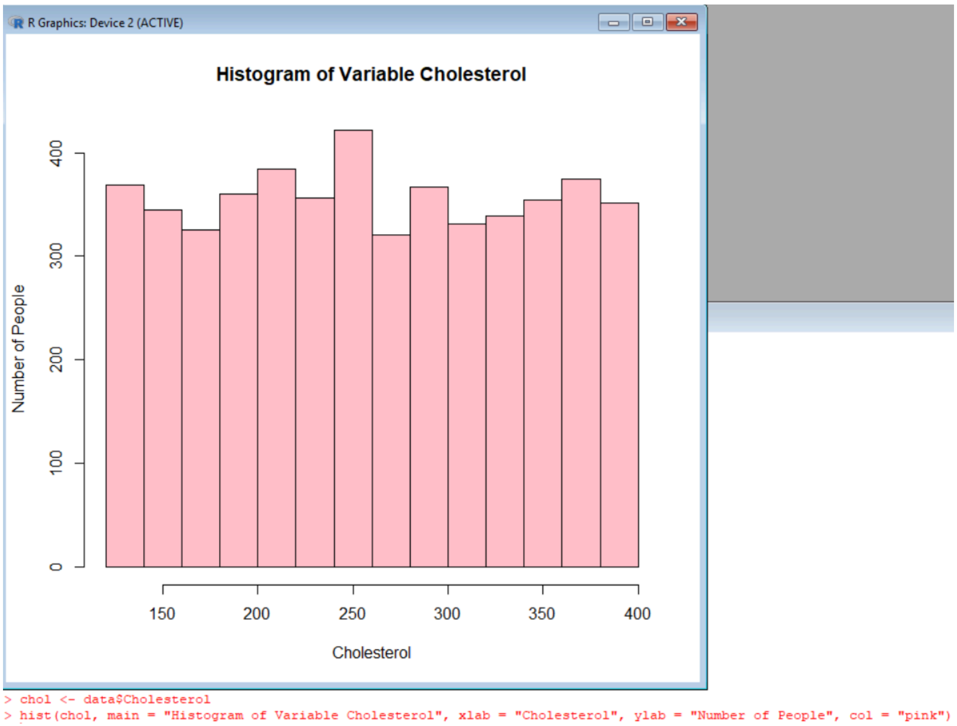
Mean	*74.8979796*
Median	*75*
Mode	*94*
Standard Deviation	*20.35493363*
Variance	*414.240442*
Range	*70*
Quartiles	*Q1: 57* *Q2: 75* *Q3: 93*
IQR	*36*
CV	*27.18%*

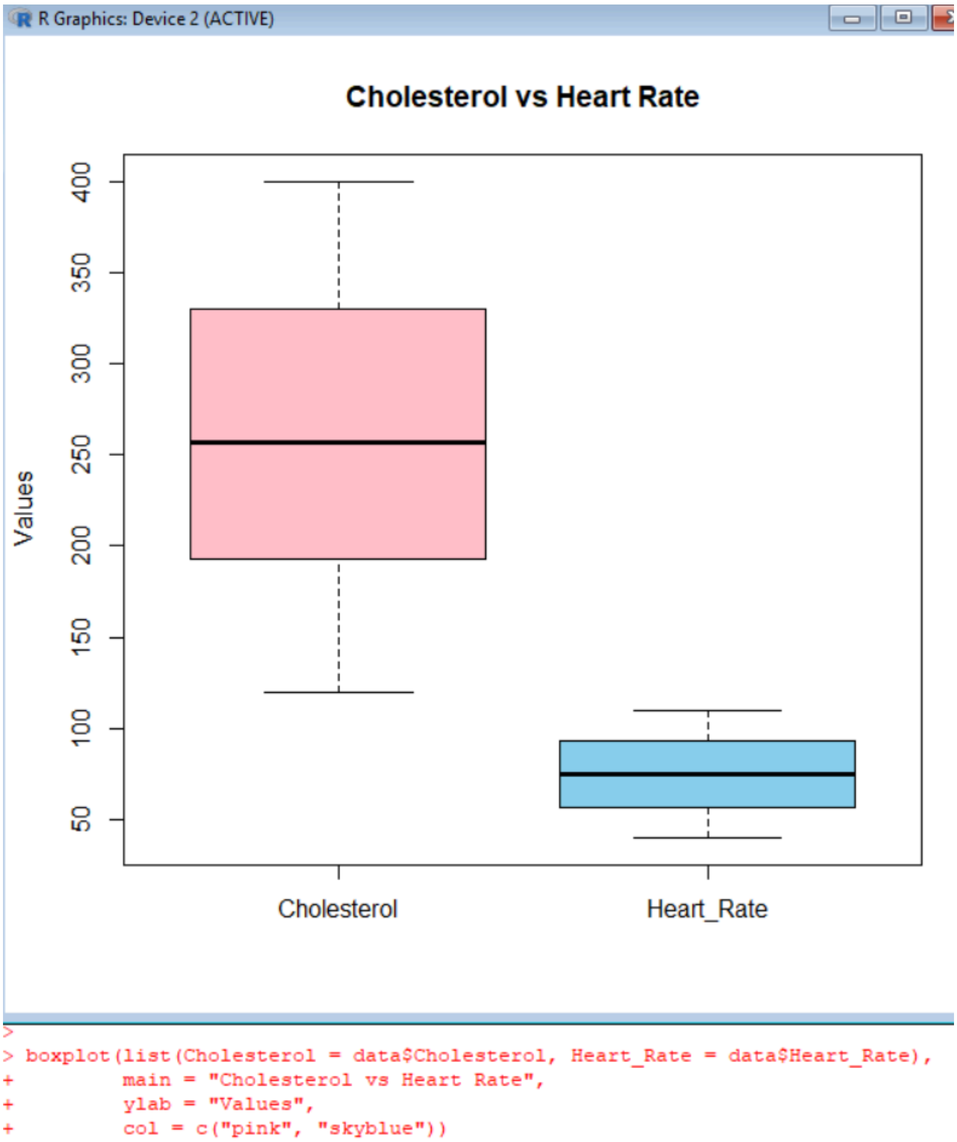
2. Compare and contrast the two variables. Discuss which is more consistent and why, using CV and standard deviation.
Heart rate is more consistent than cholesterol as it has lower CV.A CV reflects the ratio of the standard deviation to the mean ,so a smaller CV indicates more consistent and less variable measurements.

3. Use visualizations (boxplot, histogram) in R to support your explanation.
Interpret skewness, outliers and symmetry.

<table>
<tr>
<td>Heart rate histogram</td>
<td>


Read the CSV file from the specified path and store it into data frame called ‘data’
> data <-
read.csv("C:/Users/User/Downloads/Assignment_1_Statistical/heart_attack_prediction_dataset.csv")
# Extract the ‘Heart_Rete’ column and store it in variable ’heart’
> heart <- data$'Heart_Rate'
Create histogram of ‘heart’,use color sky blue for the bar
> hist(heart, main = "Histogram of Variable Heart Rate", xlab = "Heart Rate", ylab ="Number of People", col = "skyblue")
</td>
</tr>
</table>

Cholesterol histogram	 **# Extract the ‘Cholesterol’ column and store it in variable ’chol’** > chol <- data$Cholesterol **# Create histogram of ‘chol’,use color pink for the bar** > hist(chol, main = "Histogram of Variable Cholesterol", xlab = "Cholesterol", ylab = "Number of People", col = "pink")

<table>
<tr>
<td>Boxplot of Cholesterol and Heart Rate</td>
<td>


#Create a boxplot comparing Cholesterol and Heart Rate, set cholesterol as pink and heart rate as sky blue
> boxplot(list(Cholesterol = data$Cholesterol, Heart_Rate = data$Heart_Rate),
+ main = "Cholesterol vs Heart Rate",
+ ylab = "Values",
+ col = c("pink", "skyblue"))
</td>
</tr>
<tr>
<td>Explanation</td>
<td>
The median cholesterol level is lower than the mean, indicating a positively skewed distribution. This is further supported by the boxplot, where the upper quartile exceeds the lower quartile and the whiskers are longer on the upper end. The boxplot of cholesterol shows slight positive skewness. The histogram is widely spread , appears irregular and not symmetrically distributed.

In contrast, the boxplot of heart rate appears more symmetric, with the median centrally located within the interquartile range. The histogram for heart rate is generally uniform across higher values but shows a significant left-side outlier, causing the overall
</td>
</tr>
</table>

	distribution to be slightly right skewed. *Thus, heart rate is more consistent than cholesterol, as shown its boxplot shows more symmetric distribution and the histogram has only an extreme outlier on the left side, indicating more stability and less variability in the overall measurement.*

Part B: Distributional Assessment (6 marks)

1. Select one of the two variables you analysed in Part A.

   Heart Rate

2. Examine whether the variable follows a normal distribution by applying the empirical rule (68%-95%-99.7%).Calculate the proportion of data points that fall within one, two, and three standard deviations from the mean.

Calculate the intervals if the variable follows a normal distribution

Interval	*Expected range*	*Calculated Range*
$\mu \pm \sigma$	*68%*	*[54.54304596, 95.25291323]*
$\mu \pm 2\sigma$	*95%*	*[34.18811233, 115.6078469]*
$\mu \pm 3\sigma$	*99.7%*	*[13.83317869, 135.9627805]*

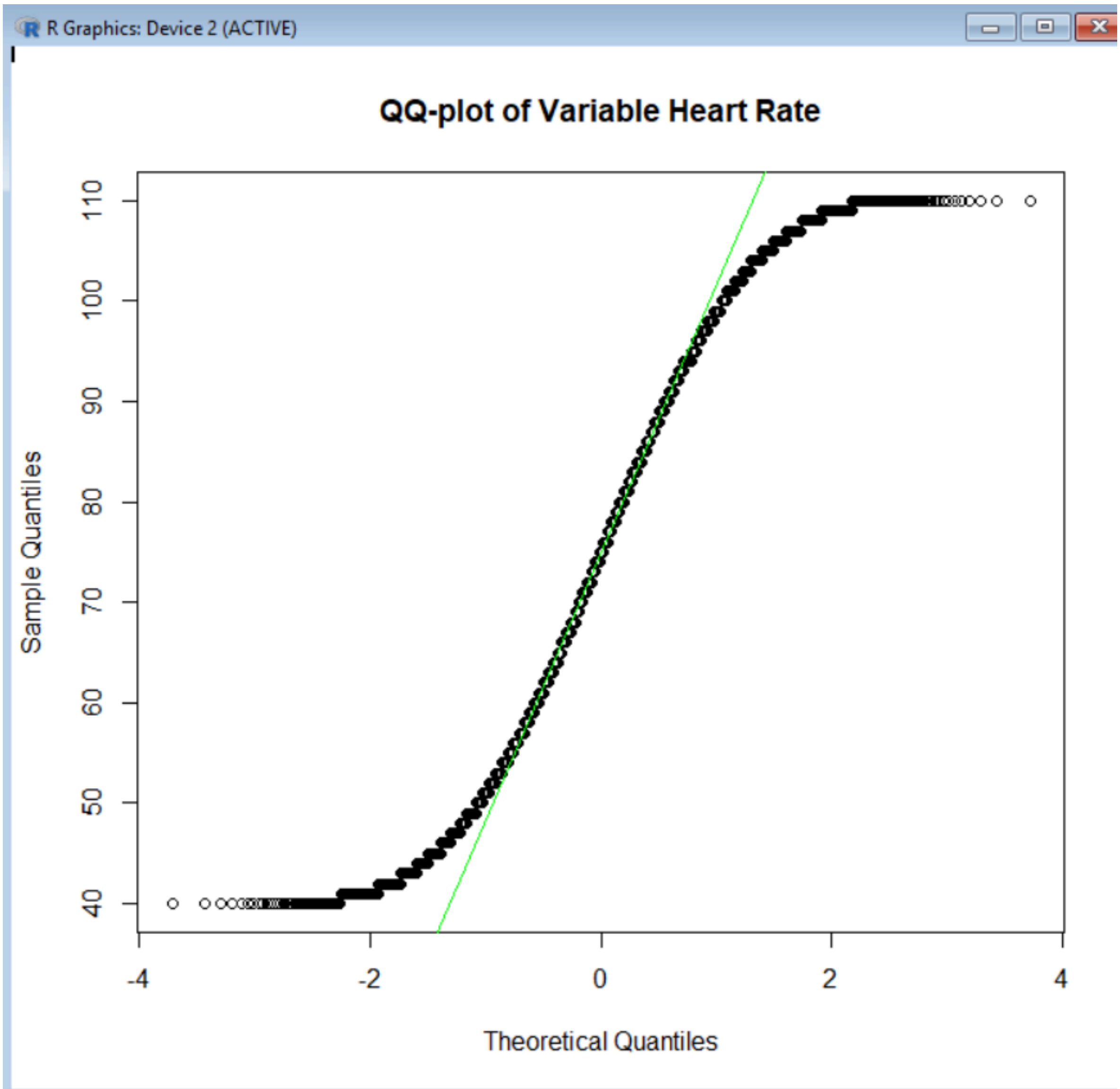
Count how many actual values fall in the intervals

Calculated Range	*Expected %*	*Actual count*	*Total count*	*Actual %*
[54.54508196, 95.25087723]	*68%*	*2926*	*5000*	*58%*
[34.19218433, 115.6037749]	*95%*	*4999*	*5000*	*99%*
[13.8392867, 135.9566725]	*99.7%*	*4999*	*5000*	*99%*

Only 58% of the data falls within $\mu \pm \sigma$, while a normal distribution would expect 68%. 99% of data falls within $\mu \pm 2\sigma$ while a normal distribution would expect 95%. 99% of data falls within $\mu \pm 3\sigma$ while a normal distribution would expect 99.7%.

$\mu \pm \sigma$ contains less data than expected, $\mu \pm 2\sigma$ and $\mu \pm 3\sigma$ contain more than expected Thus, the variable is not following a perfect normal distribution by applying the empirical rule.

3. Create a QQ-plot for the same variable using R. Then, describe whether the variable appears to be normally distributed, providing justification based on the empirical rule, QQ-plot and Shapiro-Wilk test.




```
> qqnorm(heart, main = "QQ-plot of Variable Heart Rate")
> qqline(heart, col = "green")
```

#Create Q-Q plot to check if 'heart' follow a normal distribution, by adding a reference line to indicate where the points would fall if the data is normal distributed

> qqnorm(heart, main = "QQ-plot of Variable Heart Rate")
> qqline(heart, col = "green")

```
> shapiro.test(heart)

        Shapiro-Wilk normality test

data:  heart
W = 0.9557, p-value < 2.2e-16
```

> shapiro.test(heart)

Based on empire rule, $\mu \pm \sigma$ contains less data than expected, while $\mu \pm 2\sigma$ and $\mu \pm 3\sigma$ contain more than expected. The variable does not follow a perfect normal distribution by applying the empirical rule.

Since the p- value is much less than 0.05 in the Shapiro-Wilk test, we reject the null hypothesis of the Shapiro-Wilk test. Therefore, the variable does not appear to be normally distributed based on the Shapiro-Wilk test.

The QQ-plot shows that the data points, particularly in the tails and extremes, deviating from the diagonal line. This indicates that the distribution of heart rate diverges from a normal distribution.

The variable heart rate does not follow a normal distribution.

4. Justify which method is most reliable and practical for your data.

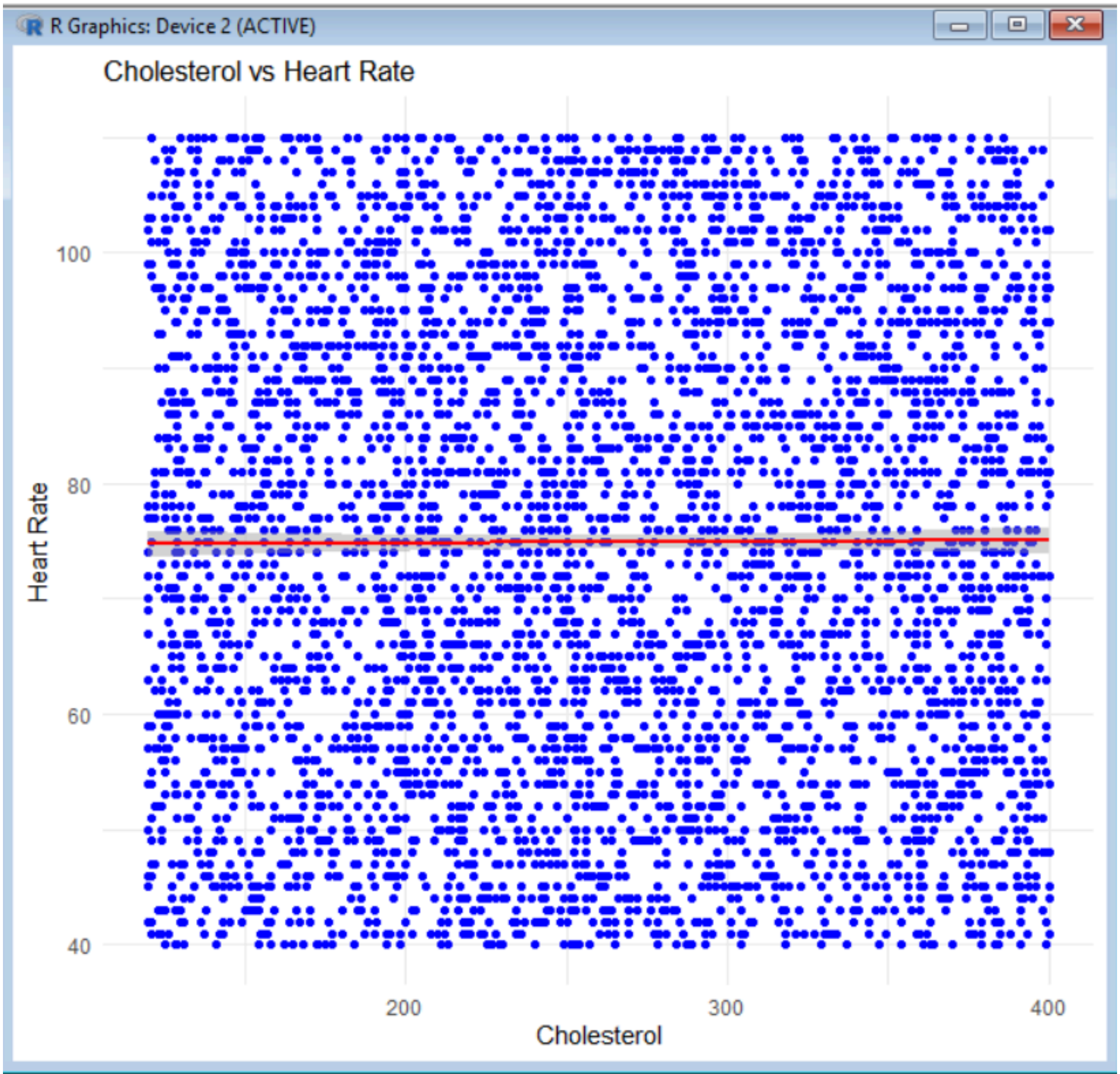
The Shapiro-Wilk test provides a p-value that clearly indicates whether the data follows a normal distribution but it is sensitive to large datasets, often rejecting normality even when the deviation is minor. In RGui, the Shapiro-Wilk test cannot be run while the dataset has more than 5000 records. Also, it does not offer insight into the nature of the deviation from normality.

The empirical rule is simple and easy to apply, making it a quick method to check for normality. However, it relies on assumptions of a symmetric and bell-shaped distribution. This makes it only reliable when the data is already approximately normal, which is rarely the case in real-world datasets.

The Q-Q plot is the most reliable and practical method for my data. It allows for visual inspection of deviations from normality and makes it easy to detect issues such as deviations in the tails and extremes, which might be missed by statistical tests. Moreover, the Q-Q plot is easy to generate for datasets of any size and provides a clear picture of how the data diverges from a normal distribution.

Part C: Bivariate Relationship Analysis (8 marks)

1. Create a scatter plot with a regression line using R to visualize the relationship between the two variables you chose in Part A.




```
> library(ggplot2)
>
> ggplot(data, aes(x = Cholesterol, y = Heart_Rate)) +
+   geom_point(color = "blue") +
+   geom_smooth(method = "lm", se = TRUE, color = "red") +
+   labs(title = "Cholesterol vs Heart Rate",
+        x = "Cholesterol",
+        y = "Heart Rate") +
+   theme_minimal()
`geom_smooth()` using formula = 'y ~ x'
```

#Load ggplot2 library for plotting
> library(ggplot2)

#Create scatter plot with regression line to show the relationship between cholesterol and heart rate
> ggplot(data, aes(x = Cholesterol, y = Heart_Rate)) +
+ geom_point(color = "blue") + **# Add blue scatter plot points**
+ geom_smooth(method = "lm", se = TRUE, color = "red") + **# Add a red linear**

regression line with confidence interval (show in shaded area) + labs(title = "Cholesterol vs Heart Rate", + x = "Cholesterol", + y = "Heart Rate") + + theme_minimal()
No correlation scatter plot with a regression line

2. Calculate the Pearson correlation coefficient (r) using the formula involving S_{xx}, S_{yy}, and S_{xy} and r^2 by using R. Interpret in context.

formula

```
> x <- data$Cholesterol
> y <- data$Heart_Rate
>
>
> x_bar <- mean(x)
> y_bar <- mean(y)
>
>
> Sxx <- sum((x - x_bar)^2)
> Syy <- sum((y - y_bar)^2)
> Sxy <- sum((x - x_bar) * (y - y_bar))
>
>
> r <- Sxy / sqrt(Sxx * Syy)
> r_squared <- r^2
>
>
> cat("Pearson r =", round(r, 4), "\n")
Pearson r = 0.0044
> cat("R-squared =", round(r_squared, 4), "\n")
R-squared = 0
```

 # Extract the 'Cholesterol' column and store it in variable 'x'
> x <- data$Cholesterol
 # Extract the 'Heart_Rate'column and store it in variable 'y'
> y <- data$Heart_Rate

#Calculate the means of heart rate and cholesterol
> x_bar <- mean(x)
> y_bar <- mean(y)

#Calculate the sum of squares for heart rate and cholesterol
> Sxx <- sum((x - x_bar)^2)
> Syy <- sum((y - y_bar)^2)

#Calculate the sum of cross-products
> Sxy <- sum((x - x_bar) * (y - y_bar))

#Calculate the Pearson correlation coefficient as ‘r’ > r <- Sxy / sqrt(Sxx * Syy) **#Calculate R-squared as ‘r_squared’** > r_squared <- r^2 **#Print out the Pearson correlation coefficient and R-square values, rounded to 4 decimal places** > cat("Pearson r =", round(r, 4), "\n") > cat("R-squared =", round(r_squared, 4), "\n")
Interpretation: *The result is Pearson correlation coefficient, Pearson r = 0.0044, while r_quared = 0.* *Pearson r = 0.0044 indicates a weak positive linear relationship between heart rate and cholesterol.The R-squared value = 0 indicates that heart rate and cholesterol do not explain any of the variability in the dependent variable.* *Thus, cholesterol and heart rate have nearly no correlation in the dataset. Changes in one variable do not predict changes in the other.*

4. Evaluate whether the relationship is causal or associative.

The relationship between cholesterol and heart rate is neither strong nor statistically significant as r=0.0044. So the relationship is neither casual nor associative.

Part D: Application G Critical Reflection (6 marks)

1. Summarize your insights for a non-technical audience (max 200 words).

This assignment aimed to explore the relationship between cholesterol, heart rate and heart attack risk. Cholesterol and heart rate are the quantitative variables, while heart attack risk is a categorical variable.

Cholesterol is unevenly spread, showing more people having higher levels than average, suggesting some individuals are at higher risk. Heart rate is more consistent across people, with fewer extreme values.

Heart rate shows a more stable pattern but neither one stands out as a strong predictor on its own. The histogram confirms that heart rate is not normally distributed which means that it does not behave in the way many health models assume.

QQ plot shows that variables deviate from what is expected in a normal distribution. There is no connection between cholesterol and heart rate, indicating they vary independently.

Thus, neither heart rate nor cholesterol can predict heart attack in this dataset. They do not show a meaningful relationship with each other. There might be more complex factors in predicting heart attack risk.

2. Discuss one limitation of descriptive statistics and how inferential stats could help.

Descriptive statistics only summarize and describe the features of a dataset, such as *the average, range and standard deviation, making it impossible to generalize findings to a larger population. Inferential statistics address this by allowing researchers to make predictions and draw conclusions about a population based on sample data in order to enhance the applicability of their findings.*

3. Reflect on how visualizations enhance understanding beyond numbers.

Visualizations of graphs can convey information at a glance, allowing researchers to understand relationships and comparisons between variables quickly. For example, the boxplot of heart rate is more symmetric, indicating that heart rate is more consistent than cholesterol. Besides that, visualization can effectively illustrate the relationship between two variables, helping to clarify how different variables interact with one another. These visualizations also help in supporting clearer communication of insights, it can help non-technical audiences to understand patterns, trends and anomalies without complex statistical formulas.

Submission Guidelines:

- Submit a PDF file including your R code, outputs, and written explanations for each part.

- Label each section clearly.

- Include your name, student ID in this format (*STUDENTID_NAME_Assignment1*).

- Plagiarism or unmodified AI outputs will definitely receive **zero marks.**